HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 215: Data Structures

Batch: 28th

Semester: Spring 2026

Date: 01/09/2026

Final Term Examination

Time: 2 Hours
Total Marks: 40

Answer any **five** questions. $5 \times 8 = 40$

1)

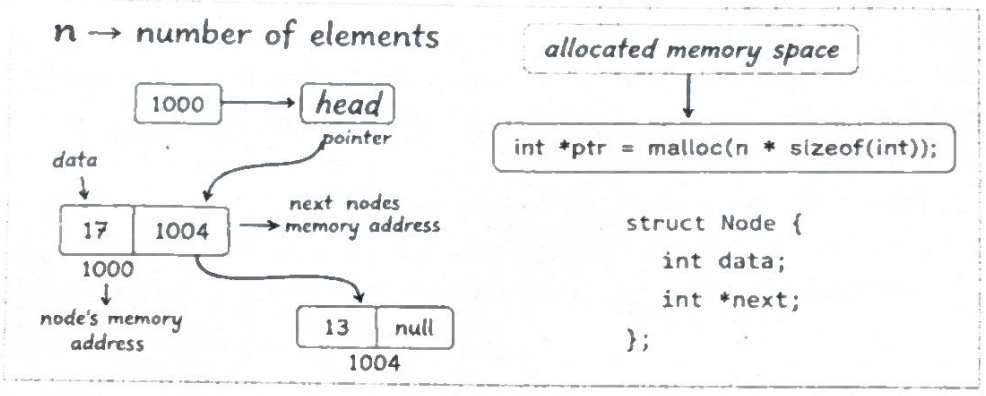


Figure 1

a. Describe the memory allocation system in ***Figure 1.*** CLO1 3

b. Illustrate the real life application of the data structure shown in ***Figure 1.*** CLO4 3

c. From ***Figure 1,*** validate the implementation of "Node" structure. CLO5 2

2)

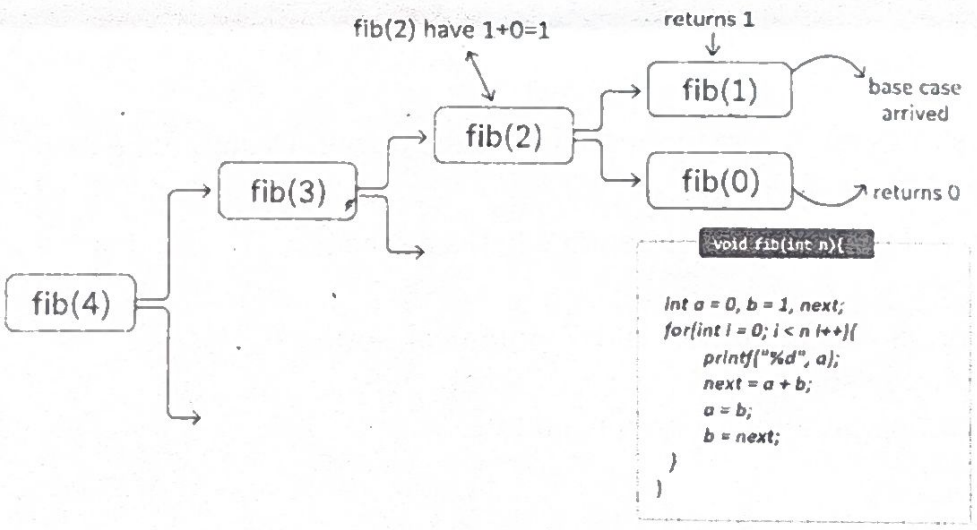


Figure 2

a. Determine whether the implemented "fib" function satisfies the theorem demonstrated by the tree in ***Figure 2.*** CLO3 3

b. From ***Figure 2***, sketch the recursive tree properly and determine the value of fib(4). CLO3 5

3) A university stores the following student marks in an array. The course teacher wants to arrange the marks in ascending order using sorting algorithms.

Student	A	B	C	D	E	F	G
Marks	71	17	7	5	3	31	13

Table 1

a.	Apply Merge Sort to arrange the marks in ascending order. Show the major steps of splitting and merging.	CLO2	4
b.	Apply Quick Sort to the same data using the first element as the pivot for sorting in descending order. Show the partitioning steps.	CLO2	4

4)

Scenario 1: A logistics company operates delivery hubs across a city and needs to know the shortest travel distance between every pair of hubs. Since the company frequently plans deliveries between different starting and destination points, the system needs a complete shortest-distance table for all locations. A graph is created where each hub is a vertex and each road is a weighted edge. The company wants to determine the shortest distance from every hub to every other hub, rather than finding a route from only one selected starting point.

a.	Define an algorithm that can be used in ***Scenario 1.***	CLO4	2
b.	Summarize your reason for choosing the algorithm in 4(a).	CLO4	3
c.	Explain the pseudo code for the algorithm you preferred previously.	CLO5	3

5)



Figure 3

a.	***Figure 3*** illustrates an important operation of the queue. Describe the real life application of the operation.	CLO5	4
b.	Define Stack and Queue and differentiate between them.	CLO1	4

6)

a.	Compare Merge Sort, Quick Sort and Topological Sort.	CLO2	3
b.	Explain the suitable scenarios of using BFS and DFS.	CLO4	3
c.	Describe the basic operation of graph and tree.	CLO5	2

7)

Event List:

i. A developer wants to develop a music playlist that continuously plays the songs in the list and does not stop until the user chooses to turn it off.

ii. Foodpanda wants an efficient algorithm to find the shortest paths between the delivery person, customer, and restaurant.

iii. A hospital management system uses a data structure to manage patient appointments such that the patient who registers first receives the doctor's appointment first.

iv. A student uses a graph to search for elements in a list. The graph is traversed level by level.

Possible Datastructures: Array, Linked List, BFS, DFS, Floyd Warshall, Dijkstra, Stack, Queue, Graph, Tree.

a.	Choose the most suitable data structure for each of the scenarios listed above, and provide a brief justification for your choice.	CLO4	8

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering

Program: BSc in CSE
Final Examination- Spring 2026
Course Title: Financial and Managerial Accounting
Course Code: ACT-215

Total Marks: 40 **Time: 2 Hours**

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answer to questions should be neat and clean.
- Answer any **Four (04)** of the following questions.

<table>
<tr><th>No.</th><th>Question/ Statement</th><th>CLOs</th><th>Marks</th></tr>
<tr><td>1.</td><td>a) List the major elements of product cost.
b) "The variable Cost per unit varies with output, whereas the fixed cost per unit is constant. "Do you agree? Explain.
c) Mannerman Fabrication manufactures a variety of products in its factory. Data for the most recent month's operations appear below:
<table>
<tr><td>Beginning raw materials inventory</td><td>55,000</td></tr>
<tr><td>Ending raw materials inventory</td><td>65,000</td></tr>
<tr><td>Direct labor</td><td>2,15,000</td></tr>
<tr><td>Purchases of raw materials</td><td>4,40,000</td></tr>
<tr><td>Manufacturing overhead</td><td>3,80,000</td></tr>
<tr><td>Beginning work in process inventory</td><td>1,90,000</td></tr>
<tr><td>Ending work in process inventory</td><td>2,20,000</td></tr>
</table>
Required:
Prepare a schedule of cost of goods manufactured for the company for the month</td><td>2

6</td><td>2
3
5</td></tr>
<tr><td>2</td><td>a) Define Mixed cost with appropriate example.
b) The Lakeshore Hotel's guests days following information for six months is obtained from a company:
<table>
<tr><th>Month</th><th>Guest-Days of Occupancy</th><th>Custodial Supplies expense</th></tr>
<tr><td>March</td><td>4,000</td><td>7,500</td></tr>
<tr><td>April</td><td>6,500</td><td>8,250</td></tr>
<tr><td>May</td><td>8,000</td><td>10,500</td></tr>
<tr><td>June</td><td>10,500</td><td>12,000</td></tr>
<tr><td>July</td><td>12,000</td><td>13,500</td></tr>
<tr><td>August</td><td>9,000</td><td>10,750</td></tr>
<tr><td>September</td><td>7,500</td><td>9,750</td></tr>
</table>
Required:
i. Ascertain variable cost per unit and total fixed cost under the High and Low Method.
ii. Using the cost formula ,what amount of custodial supplies expense would you expect to be incurred at an occupancy level of 11,000 guest-days?</td><td>6</td><td>2
8</td></tr>
</table>

6000

-1.33

$y = a + bx$

3.	a) Discuss in brief different types of Adjusting entries. b) The Meghna Lawn care company Ltd. began on April 1. At April 30, the trial balance shows the following balances for selected accounts (Amounts are in Taka). Prepaid Insurance 3,600 Equipment 28,000 Notes Payable 20,000 Unearned Service Revenue 4,200 Service Revenue 1,800 Analysis reveals the following additional data. 1. Prepaid insurance is the cost of a 2-year insurance policy, effective April 1 2. Depreciation on the equipment is Tk.500 per month. 3. The note payable is dated April 1. It is a 6-month, 12% note. 4. Seven customers paid for the company's 6-month lawn service package of Tk.600 beginning in April. The company performed services for these customers in April. 5. Lawn services performed for other customers but not recorded at April 30 totaled Tk.1,500. **Instructions:** Prepare the adjusting entries for the month of April. Show computations	4	3 7
4	a) Interpret the term CVP analysis. b) Menlo company distributes a single product. The company's sales and expenses for last month Total (Tk.) / Per Unit (Tk.) Sales 450,000 30 Variable expenses 180,000 12 Contribution margin 270,000 18 Fixed expenses 216,000 Net operating Income 54,000 i. Compute the company's CM ratio and Variable Cost ratio. ii. What is the monthly break-even point in units sold and in sales dollars? iii. How many units would have to be sold each month to earn a target profit of $90,000? Use the formula method. Verify your answer by preparing a contribution format income statement at the target sales level. iv. Refer to the original data. Compute the company's margin of safety in both dollar and percent- age terms. v. Compute the company's operating leverage	3,6	3 7
5	a) Differentiate between absorption costing and variable costing. b) Maxwell Company manufactures and sells a single product. The following costs were incurred during the company's first year of operations: **Variable costs per unit: Manufacturing:** Direct materials 18 Direct labor 7 Variable manufacturing overhead 2 Variable selling and administrative 5 **Fixed costs per year:** Fixed manufacturing overhead 1,60,000 Fixed selling and administrative expenses 1,10,000 During the year, the company produced 20,000 units and sold 16,000 units. The selling price of the company's product is $50 per unit. Required: Assume that the company uses absorption costing: i. Compute the unit product cost. ii. Prepare an income statement for the year.	3,6	3 7
6	Write short Note (Any five); a) Financial Accounting vs Managerial Accounting. b) Contribution margin and Margin of safety. c) Sunk cost with example. d) Prime cost and Conversion cost with example. e) Product cost and period cost with example. f) Accrual basis accounting and Cash basis accounting g) Break-even point (BEP).	2,3,4,6	10

HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
Program: B.A. (Hons)
3rd Semester Examination

Course Code: BAN211
Course Title: History of Emergence of Bangladesh
Final Examination
Date: 06.09.2026 Total Time: 2 Hours Semester: Spring 2026
Total Marks: 40

Instruction & Information

- **Answer any *Four* questions**
- **All questions carry equal marks**

1. Discuss the process of Government. formation of Bangladesh(1971 to 2026). 10

2. How many times has the constitution of Bangladesh been amended? Mention the reason with explanation. 10

3. Discuss organs of Bangladesh Government. 10

4. Discuss the duties and responsibilities of a Citizen to the State. 10

5. Discuss how you would serve the Country as a Citizen. 10

HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science (Honours) in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213 **Course Title: Statistics & Probability**
Final Examination Semester: Spring 2026
Date: 9/6/2026 Total Time: 2:00 Hours **Total Marks: 40**

Answer any four (04) out of the following questions: **$4 \times 10 = 40$**
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No		Question	CLOS	Marks
Q1.	(a)	Define central tendency. Also explain different types of central tendency.	CLO1	[3]
	(b)	Write down the characteristics of central tendency.	CLO2	[2]
	(c)	A factory has 150 workers. The monthly income (in taka) of 10 1workers selected randomly from this factory are as follows: 344, 347, 356, 349, 357, 367, 345, 384,372. Then show that $A.M \geq G.M \geq H.M.$	CLO3	[5]
Q2.	(a)	Write down the merit and demerits of mean, median, and mode.	CLO1	[2]
	(b)	Calculate the median from the given data sets are as follows: 2500, 2820, 2640, 2570, 2780, 2610, 2760, 2840,2700, 2670.	CLO3	[2]
	(c)	The following frequency distribution represents the marks obtained by 50 students in a Mathematics examination.	CLO3	[6]

Class interval	Frequency
0-10	5
10-20	8
20-30	12
30-40	15
40-50	7
50-60	3

$Me = L + \frac{\frac{N}{2} - F}{f} \times h$

(i) Calculate the median of the distribution using the approach formula.
(ii) Construct a less-than cumulative frequency table and determine the median graphically.

Q3. **(a)** Define Arithmetic Mean, Geometric Mean and Harmonic Mean. Among these three measures which one is considered the best and why? CLO1 [4]

(b) The following table shows the number of units sold by a retail shop at difference size of shoes. CLO3 [6]

Size of shirts	39-40	40-41	41-42	42-43	43-44
No. of pair sold	9	13	12	8	18

Using the appropriate formula to determine the mode of the selling number of shoes and also locate it graphically.

Q4. **(a)** Write down a short note on difference types of set. CLO1 [3]

(b) Suppose $P[L] = 0.25, P[M] = 0.45, P[L \cap M] = 0.15$ then CLO3 [4]

(i) Are L and M independent and mutually exclusive?

(ii) Find $P[L/M] = ?$

(iii) State the nature of the events L and M.

(c) A box contains 36 items which five items are defective. An item as selected at random from the box. Find the probability that the selected item- CLO3 [3]

(i) Non-defective

(ii) Defective or non-defective.

Q5. **(a)** What is a quartile? Distinguish between first quartile (Q_1) and third quartile (Q_3). CLO1 [2]

(b) The following table shows the number of hours worked per month by a group of workers: CLO3 [8]

No. of hours worked per month	30-55	55-80	80-105	105-130	130-155
No. of workers	3	4	12	6	9

(i) Calculate the Arithmetic Mean of the distribution.

(ii) Calculate Median Q_1, D_3, and P_{70}.

(iii)Locate Median, Q_1, D_3, and P_{70}.

$$Li + \frac{\frac{n}{2} - Fi}{Fi} \times ci$$

Q6. **(a)** Define probability. Graphically represent dependent, independent, mutually exclusive and not mutually exclusive events. CLO1 [3]

(b) In a class of 100 students 90 took CSE and took 85 Mathematics and 65 took both. A student is selected at randomly from this class. Calculate the probability that selected student took – CLO3 [7]

(i) At least one course.

(ii) Both the course.

(iii)Only CSE.

(iv)Only one course.

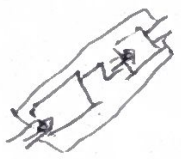
NAMOORD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE213: Digital Electronics

Batch: 28th | Semester: Spring 2026
Date: 10/9/2026 | **Final Examination** | Time: 2 Hours
Total Marks: 40

Answer any **five** questions of the followings. 5 × 8 = 40

1.	a)	Explain Half adder with proper diagram and truth table.	CLO1	3
	b)	What is Full adder? Implement Full adder with Half adders.	CLO1	5
2.	a)	Do you support the statement "T flip-flop has an invalid state"? Justify your answer while providing all the tables and equations regarding T flip-flop.	CLO4	4
	b)	Differentiate between latch and flip-flop.	CLO4	4
3.	a)	Define the following terms i) Synchronous counter ii) Asynchronous counter	CLO4	2
	b)	Explain SR flip flop with necessary tables and equations.	CLO4	6
4.	a)	Design a 2-bit asynchronous down counter.	CLO4	2
	b)	Explain race condition in JK flip flop with proper timing diagram. Describe the remedy of race condition.	CLO4	6
5.	a)	Design a modulus-3 asynchronous upcounter.	CLO4	3
	b)	Design a 3-bit asynchronous up-down counter.	CLO4	5
6.		Explain JK flip flop with necessary tables and equations.	CLO4	8
7.		Design a 2-bit synchronous up-down counter.	CLO4	8



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science (Honours) in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT217 **Course Title: Ordinary and Partial Differential Equations**
Final Examination Semester: Spring 2026
Date: 9/13/2026 Total Time: 2:00 Hours **Total Marks: 40**

Answer any four (04) out of the following questions: **$4 \times 10 = 40$**
[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

No		Question	CLOS	Marks
Q1.	(a)	Define partial differential equation and also classify different types of partial differential equations with example.	CLO1	[3]
	(b)	Eliminating constants a, b, c from the partial differential equations:	CLO3	[7]
		(i) $z = a(x + y) + b.$		
		(ii) $\ln(az - 1) = x + ay + b.$		
		(iii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$		
Q2.	(a)	Define semi-linear and quasi-linear partial differential equation with example.	CLO1	[2]
	(b)	From a partial differential equation of a family of a planes ,the sum of whose x, y, z intercepts is equal to zero.	CLO2	[4]
	(c)	From a partial differential equation by eliminating arbitrary function from the following relation $\emptyset(x + y + z, x^2 + y^2 - z^2) = 0.$	CLO3	[4]
Q3.	(a)	Define the order and degree of a partial differential equation. Determine the order, degree, and linearity of $\left[2 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{1}{2}} = m\frac{d^3y}{dx^3}.$	CLO1	[3]
	(b)	Solve the following partial differential equations:	CLO3	[7]
		(i) $z = ae^{bt} \sin bx.$		
		(ii) $(x^2 - y^2 - yz)\, p + (x^2 - y^2 - yz)\, q - z(x - y) = 0.$		
Q4.	(a)	State and prove Lagrange's method of solving the linear partial differential equation of order one namely $Pp + Qq = R, p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial x}$ and P, Q, R are the function of $x, y, z.$	CLO2	[6]
	(b)	Solve the partial differential equation by using Lagrange's method: $(3x + y - z)p - 2(z - y) = -(x + y - z)q.$	CLO3	[4]
Q5.	(a)	Describe integral surface with example.	CLO1	[2]
	(b)	Find the system of the surfaces which intersect the following surface of the system orthogonally.	CLO3	[4]
	(c)	Evaluate the integral surfaces of the partial differential equation: $2y(z - 3)p + (2x - z)q = y(2x - 3), x^2 - 2x = -y^2, z = 0.$	CLO3	[4]

Q6. **(a)** State and prove Charpit's method for solving first-order nonlinear partial differential equation. CLO2 [5]

(b) Apply charpit's method to find a complete integral of the following differential equations: CLO3 [5]

(i) $zpq = p + q$.

(ii) $p^2 - y^2 = y^2q - x^2$.